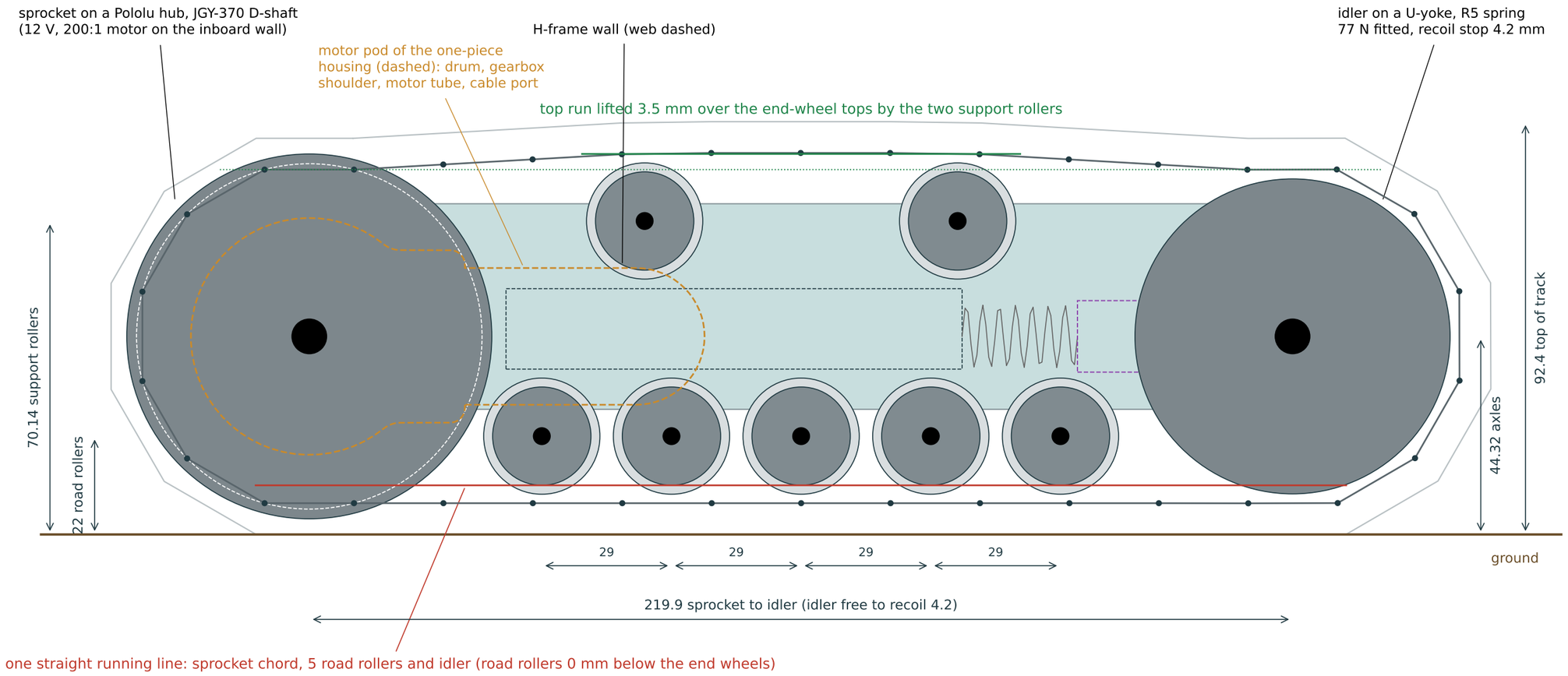
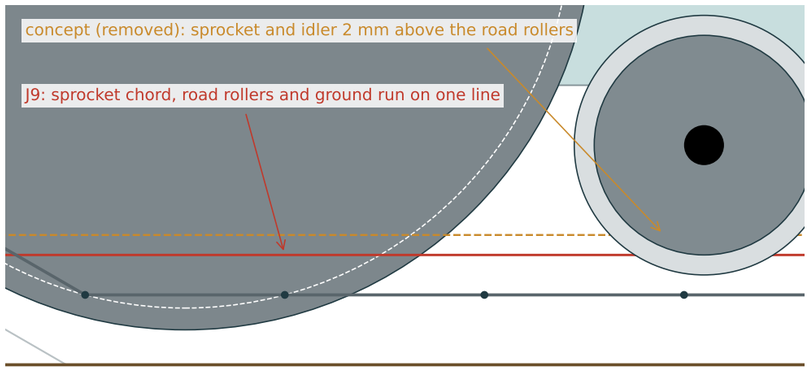
JGY-370 single track — J9 layout (built from the CAD data)

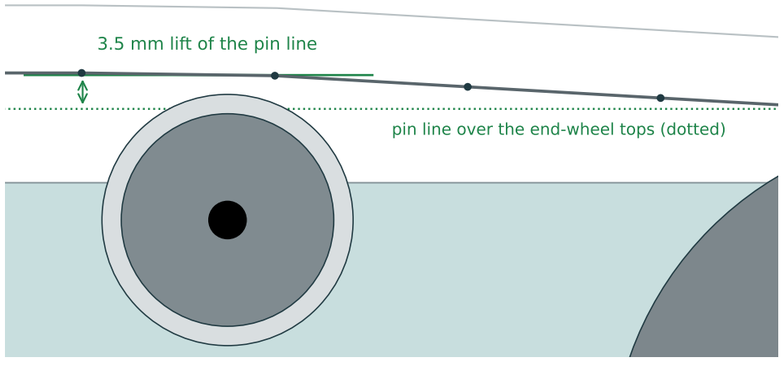
Change after the concept review: flat ground run. Sprocket, five road rollers and idler bear on one line; only the two support rollers lift the track.
Side view from the inboard (motor) side with the housing and motor hidden; the housing's motor pod is dashed. All dimensions in mm.



DETAIL A — sprocket end of the ground run



DETAIL B — support roller lifting the top run



	concept	J9 CAD
Road rollers below end wheels	2.0	0 (flat)
Axle height	46.32	44.32
Idler centre X (34 links taut)	109.80	109.90
Support-roller lift / centre Z	3.5 / 72.14	3.5 / 70.14
Top of track	94.4	92.4
Sprocket outer support	boss in 6800ZZ	plug in 6800ZZ
Idler axle	M8 bolt, wall slots	Ø8 pin in yoke
Sprocket on the motor shaft	printed D-bore	Pololu #1999 hub
JGY-370-1260 12 V ratio	open	200:1, 30 rpm
Motor enclosure	thin shroud	closed drive pod
Spring preload / at recoil stop	77 N	77 / 104 N

CAD checks (validation.json): PASS

34 links closed exactly (error 2e-14 mm); static overlap: 0 of 449 pairs
drive rotation, idler recoil to the stop, 3 chain phases, sprocket fitting path,
plug, yoke and housing insertion, 25 tool-access paths: no overlaps
min clearance track pins/clips to H-frame 3.5 mm